FASTROBOX 1.0

AI-Powered Smart Classroom: Safety, Access and Vision Challenge

Team Name: Do-Something

(Team Number: 20)

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1. Introduction

FastRobox 1.0 is a 12-level challenge to build an AI-powered smart classroom using one ESP32-S3-CAM and the provided hardware kit. The system combines room safety monitoring, fire and panic response, secure access control, attendance, a cloud dashboard, camera-based vision, Wi-Fi sensing and offline recovery. Each level extends the previous one; earlier functions must continue working as new features are added.

2. Overall System Objectives

- Monitor classroom temperature, humidity, heat index and flame conditions.
- Keep the simulated classroom door safe and apply fail-safe behaviour during fire or emergencies.
- Provide secure Wi-Fi setup, telemetry delivery and a searchable cloud record.
- Give Admin, Teacher, Student and Guardian users access only to permitted information.
- Support remote settings, firmware updates, weather notices, reports and an AI assistant.
- Verify student identity using RFID card and on-device face detection.
- Use camera vision for occupancy, live view, night watch and hand-raise detection.
- Use Wi-Fi CSI alone for presence detection and room-zone localization.
- Remain accessible during network outages and recover safely after power cuts.

3. Hardware and Main Components

ESP32-S3-CAM	**Main controller; camera, onboard RGB LED, BOOT button, Wi-Fi and Bluetooth.**
RC522 RFID reader	Reads student cards and the Teacher key tag.
DHT11 sensor	Measures classroom temperature and relative humidity.
IR flame sensor	Detects flame for fire-state activation.
Active buzzer	Provides alarm and access feedback beeps.
Jumper wires	Sensor connections and capacitive-touch panic pad.
USB cable	Programming, Serial Monitor and board power.

4. Common System Requirements

- Use one ESP32-S3-CAM only and the supplied kit; do not add another microcontroller or extra electronic modules.
- Run independent FreeRTOS tasks across both CPU cores; avoid blocking waits and keep watchdog protection active.
- Serial Monitor communication uses 9600 baud; print the [STAT] status line every 3 seconds.
- The supported Serial commands include info and threads.
- The RGB light heartbeat is 1 Hz: 500 ms ON and 500 ms OFF, while the displayed colour follows state priority.

- Device setup access point: SmartRoom-<TeamID>, available at 192.168.10.1.
- After setup, use the mDNS address classroom-<teamid>.local.
- Send telemetry every 15 seconds using camelCase JSON fields: deviceId, roomId, seq, ts, uptimeS, tempC, rh, heatIndexC, flame, door, status, heatTrend, rssi, heapKb, chipTempC, fw and quality.
- Use secure transport (HTTPS or MQTT over TLS), validate the server certificate and keep credentials out of source code and the repository.
- Preserve all previously completed level features as later levels are added.

5. Level-wise Documentation

The following sections summarize the purpose, implementation requirements and evidence expected for each level.

Level 1: Dual-Core Tasks, Health and Watchdog

Scenario: The classroom safety device must run several jobs at once and recover automatically if a task freezes.

Purpose: Establish a reliable embedded-device foundation with concurrent sensor reading, diagnostics and automatic recovery.

Main requirements

- Create separate FreeRTOS tasks on both ESP32-S3 CPU cores, with no blocking waits.
- Read DHT11 every 5 seconds, flame sensor every 100 ms and RFID reader every 200 ms; report only one card event per tap.
- Blink the RGB LED at 2 Hz in the current state colour (green in Safe state).
- Print the fixed status information, including chip temperature, free heap and free PSRAM; support help, status, health, tasks, ver and hang commands.
- Enable the task watchdog so a frozen task causes an automatic restart within 10 seconds.
- On boot, report the reset reason: POWER_ON, SOFTWARE, WATCHDOG, CRASH or BROWNOUT.
- At startup, check flame first: no flame means door LOCKED; flame means door UNLOCKED.
- Detect unplugged/failed sensors, report SENSOR ERROR, show magenta, and keep other tasks running; clear the error after reconnection.

Where the result must appear

- Serial Monitor: status, card, sensor error and boot events; RGB LED: green heartbeat or magenta sensor error.

Live judging focus

- Steady 2 Hz status light; one CARD line per tap; health and task information; watchdog restart within 10 seconds.

Video evidence

- Unplug DHT11 data and RC522 SDA for 15 seconds; show the corresponding errors, magenta light and continued operation; show recovery after reconnecting.

Write-up evidence

- Wiring diagram with every pin connection; task diagram with task name, CPU core and period; fixed status output and sensor read periods.

Table Name: Components and Pin Layouts

Part	Pin Name	ESP32-S3 WROOM pin
RC522	SDA	GPIO 1
	SCK	GPIO 41
	MOSI	GPIO 40
	MISO	GPIO 42
	RST	GPIO 3
	3.3V	3V3
	GND	GND
DHT11	DATA	GPIO 14
	VCC	3V3
	GND	GND
IR flame module	DO	GPIO 21
	VCC	3V3
	GND	GND
Active buzzer	+	GPIO 2
	-	GND
Onboard RGB LED	Built in	GPIO 48

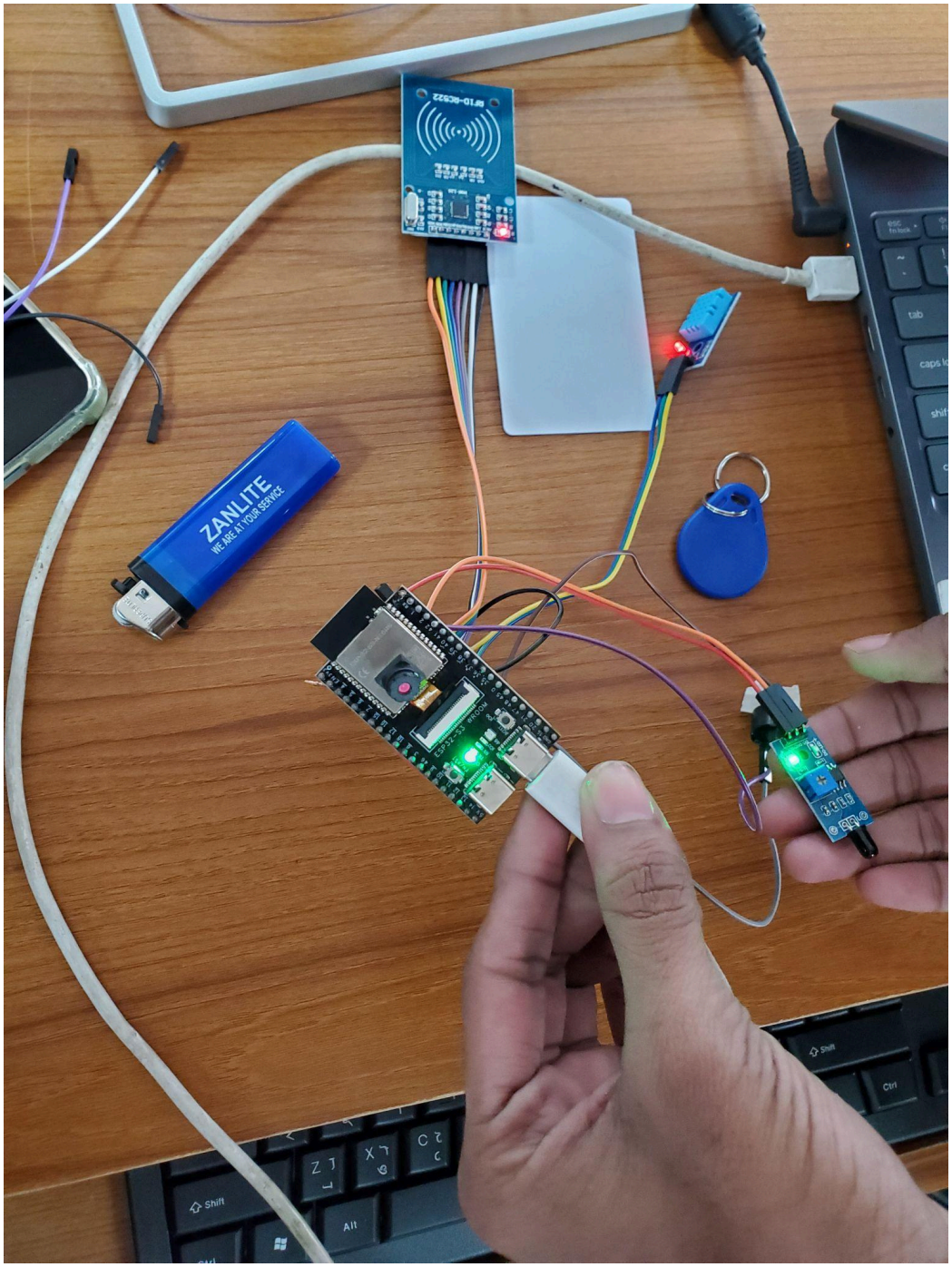


Fig: IoT Built

Outputs:

```
[STATUS] up=129s T=28.5C RH=86% HI=- flame=0 door=LOCKED state=SAFE trend=- rssi=- heap=320k psram=0k chip=30.5C fw=1.0.0
[CARD] uid=3183640A
[STATUS] up=130s T=28.5C RH=86% HI=- flame=0 door=LOCKED state=SAFE trend=- rssi=- heap=320k psram=0k chip=30.5C fw=1.0.0
[CARD] uid=979D7D33
[STATUS] up=131s T=28.5C RH=86% HI=- flame=0 door=LOCKED state=SAFE trend=- rssi=- heap=320k psram=0k chip=30.5C fw=1.0.0
```

```
[STATUS] up=28s T=27.4C RH=49% HI=- flame=0 door=LOCKED state=SAFE trend=- rssi=- heap=320k psram=0k chip=28.5C fw=1.0.0
[STATUS] up=29s T=27.4C RH=49% HI=- flame=0 door=LOCKED state=SAFE trend=- rssi=- heap=320k psram=0k chip=28.5C fw=1.0.0
[STATUS] up=30s T=27.4C RH=49% HI=- flame=0 door=LOCKED state=SAFE trend=- rssi=- heap=320k psram=0k chip=28.5C fw=1.0.0
[STATUS] up=31s T=27.4C RH=49% HI=- flame=0 door=LOCKED state=SAFE trend=- rssi=- heap=320k psram=0k chip=28.5C fw=1.0.0
[STATUS] up=32s T=27.2C RH=49% HI=- flame=0 door=LOCKED state=SAFE trend=- rssi=- heap=320k psram=0k chip=28.5C fw=1.0.0
[STATUS] up=33s T=27.2C RH=49% HI=- flame=0 door=LOCKED state=SAFE trend=- rssi=- heap=320k psram=0k chip=28.5C fw=1.0.0
[STATUS] up=34s T=27.2C RH=49% HI=- flame=0 door=LOCKED state=SAFE trend=- rssi=- heap=320k psram=0k chip=28.5C fw=1.0.0
[STATUS] up=35s T=27.2C RH=49% HI=- flame=0 door=LOCKED state=SAFE trend=- rssi=- heap=320k psram=0k chip=28.5C fw=1.0.0
[STATUS] up=36s T=27.2C RH=49% HI=- flame=0 door=LOCKED state=SAFE trend=- rssi=- heap=320k psram=0k chip=28.5C fw=1.0.0
[STATUS] up=37s T=32.8C RH=95% HI=- flame=0 door=LOCKED state=SAFE trend=- rssi=- heap=320k psram=0k chip=28.5C fw=1.0.0
[STATUS] up=38s T=32.8C RH=95% HI=- flame=0 door=LOCKED state=SAFE trend=- rssi=- heap=320k psram=0k chip=28.5C fw=1.0.0
```

```
[STATUS] up=73s T=30.0C RH=94% HI=- flame=0 door=LOCKED state=SAFE trend=- rssi=- heap=320k psram=0k chip=29.5C fw=1.0.0
[STATUS] up=74s T=30.0C RH=94% HI=- flame=0 door=LOCKED state=SAFE trend=- rssi=- heap=320k psram=0k chip=29.5C fw=1.0.0
[STATUS] up=75s T=30.0C RH=94% HI=- flame=0 door=LOCKED state=SAFE trend=- rssi=- heap=320k psram=0k chip=29.5C fw=1.0.0
[STATUS] up=76s T=30.0C RH=94% HI=- flame=1 door=UNLOCKED state=FIRE trend=- rssi=- heap=320k psram=0k chip=30.5C fw=1.0.0
[STATUS] up=77s T=30.3C RH=94% HI=- flame=1 door=UNLOCKED state=FIRE trend=- rssi=- heap=320k psram=0k chip=30.5C fw=1.0.0
[STATUS] up=78s T=30.3C RH=94% HI=- flame=1 door=UNLOCKED state=FIRE trend=- rssi=- heap=320k psram=0k chip=30.5C fw=1.0.0
[STATUS] up=79s T=30.3C RH=94% HI=- flame=1 door=UNLOCKED state=FIRE trend=- rssi=- heap=320k psram=0k chip=30.5C fw=1.0.0
[STATUS] up=80s T=30.3C RH=94% HI=- flame=1 door=UNLOCKED state=FIRE trend=- rssi=- heap=320k psram=0k chip=30.5C fw=1.0.0
```

```
ESP-ROM:esp32s3-20210327
Build:Mar 27 2021
rst:0x1 (POWERON),boot:0x2a (SPI_FAST_FLASH_BOOT)
SPIWP:0xee
mode:DIO, clock div:1
load:0x3fce2820,len:0x10cc
load:0x403c8700,len:0xc2c
load:0x403cb700,len:0x30b0
entry 0x403c88b8
[EVENT] BOOT reason=POWER_ON
[STATUS] up=1s T=- RH=- HI=- flame=0 door=LOCKED state=SAFE trend=- rssi=- heap=316k psram=0k chip=28.5C fw=1.0.0
[STATUS] up=2s T=27.6C RH=49% HI=- flame=0 door=LOCKED state=SAFE trend=- rssi=- heap=320k psram=0k chip=28.5C fw=1.0.0
```

```
[EVENT] BOOT reason=POWER_ON
[STATUS] up=1s T=- RH=- HI=- flame=1 door=UNLOCKED state=FIRE trend=- rssi=- heap=316k psram=0k chip=29.5C fw=1.0.0
[ERROR] SENSOR ERROR: RFID
[STATUS] up=2s T=0.3C RH=0% HI=- flame=1 door=UNLOCKED state=FIRE trend=- rssi=- heap=320k psram=0k chip=29.5C fw=1.0.0
[STATUS] up=3s T=0.3C RH=0% HI=- flame=1 door=UNLOCKED state=FIRE trend=- rssi=- heap=320k psram=0k chip=29.5C fw=1.0.0
[STATUS] up=4s T=0.3C RH=0% HI=- flame=1 door=UNLOCKED state=FIRE trend=- rssi=- heap=320k psram=0k chip=29.5C fw=1.0.0
[STATUS] up=5s T=0.3C RH=0% HI=- flame=1 door=UNLOCKED state=FIRE trend=- rssi=- heap=320k psram=0k chip=29.5C fw=1.0.0
```